Chapter

Dynamic Difficulty Adjustment in Games: Concepts, Techniques, and Applications

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Abstract

Modern video games face the challenge of maintaining player engagement across a wide range of skills, experiences, and motivations. Traditional fixed difficulty settings often fail to meet this need, leading to early frustration for new players and boredom for experienced ones. Dynamic difficulty adjustment provides a flexible alternative by adapting gameplay in real-time based on player performance or emotional state. Grounded in concepts such as Flow Theory and Self-Determination Theory, adaptive systems seek to align challenge with player ability and psychological needs. This chapter reviews the historical development of difficulty systems, outlines performance-based, emotion-based, and hybrid adjustment methods, and examines a case study comparing five difficulty strategies within a custom first-person shooter game. Results show that while difficulty levels and perceived stress varied across methods, players' engagement, excitement, and enjoyment remained consistent, suggesting that adaptive systems can help maintain an optimal experience across diverse player profiles. The chapter concludes by discussing the limitations of dynamic adjustment, including the importance of transparency, the risk of undermining player competence, and the need to consider different motivations for play beyond the pursuit of challenge.

Keywords: dynamic difficulty adjustment, gameplay metrics, physiological sensors, player engagement, affective computing

1. Introduction

Video games today face a delicate challenge: how to keep players engaged without overwhelming or boring them. This balance is especially tricky because players come with vastly different skill levels, gaming experience, and even daily moods. A game that's too easy might feel dull; too hard, and it quickly becomes frustrating. Traditionally, game designers address this with fixed difficulty settings—"easy," "medium," or "hard"—but these one-size-fits-all approaches rarely capture the full range of player needs.

For some players, especially those new to gaming, even the "easy" setting can feel overwhelming. They may be unfamiliar with controls, core mechanics, or genre conventions, and often do not play long enough to adapt. Many non-gamers try a game once, feel lost or frustrated, and never return. This early drop-off can prevent them from reaching the positive feedback loop of skill development, increased confidence, and eventual mastery that keeps players engaged long term. On the opposite end of the spectrum are experienced or competitive players who may find even the hardest setting insufficiently challenging, resulting in boredom rather than engagement.

Designing for these outliers—novices and experts alike—requires flexible systems that respond to individual needs. Dynamic Difficulty Adjustment, or DDA, offers a promising solution. Rather than locking players into a fixed challenge level, DDA systems adjust the game in real time based on how a player performs—or even how they feel. By tailoring difficulty to each individual, DDA can help more players find their way into the game and stay engaged, whether they are picking up a controller for the first time or seeking a new level of challenge.

In this chapter, we explore the foundations of DDA, review key approaches, including performance-based and emotion-based systems, and share insights from a study that compared several difficulty adjustment strategies. This highlights how adaptive design can shape player experience and broaden a game's reach.

2. Origins of game difficulty

The idea of adapting difficulty based on the player's performance is not new. In fact, it can be traced back to the earliest arcade games of the 1970s and 1980s. These games often featured reactive mechanics that adjusted challenge on the fly, not out of player empathy, but for commercial reasons. As players survived longer, games like Space Invaders, Tetris, and Galaga increased the speed or complexity of their enemies. These adjustments kept the gameplay from becoming monotonous and nudged players toward a quicker end, encouraging repeat plays and, ideally, another coin in the machine.

Although primitive by today's standards, these early systems introduced a key idea: challenge can and should respond to player behavior. This real-time reactivity laid the seeds for modern adaptive systems, even if it was initially driven more by monetization than by user experience.

As games transitioned from arcades to home consoles and PCs, the model shifted. Developers began offering fixed difficulty settings—Easy, Medium, Hard—as a way to give players more control. Without the pressure to push players into game-over states for profit, games could offer longer, more narrative-driven experiences. But while this new model reduced pressure, it also assumed players knew how hard the game would be and how skilled they were before even starting. That assumption would prove limiting, especially for new players or those whose skill developed significantly over time.

3. Why fixed difficulty falls short

Despite their widespread use, fixed difficulty settings introduce significant limitations for both players and developers. While they were a logical evolution from the reactive mechanics of arcade-era design, fixed settings rely on the assumption that players can accurately assess their skill level and predict how challenging a game will be before they have even begun. In reality, this often leads to mismatches: players may underestimate themselves and breeze through the game with little engagement, or overestimate and hit a wall of frustration early on.

These decisions are often made before players have a clear understanding of the situation. New players, in particular, may be unfamiliar with the game's mechanics, pacing, or difficulty curve. Being asked to choose a challenge level before even touching the controls can feel alienating, especially when that choice directly shapes the entire experience.

For developers, fixed difficulty adds another layer of complexity. Designing multiple tiers that feel distinct, fair, and fun requires a deep understanding of player psychology and broad testing across varied skill levels. Despite best efforts, these tiers are rarely one-size-fits-all. Launching with fixed difficulty tiers that are too rigid or poorly tuned can result in mismatches between player ability and challenge, leading to frustration, disengagement, or uneven reviews. Even when multiple settings are available, they may fail to adapt to players whose skill develops over time or whose preferences shift across sessions. And while post-launch patches can tweak balance, they often come too late to recover lost player interest.

Another key issue is that fixed settings do not account for player improvement. As players progress, they often become more skilled, mastering mechanics, recognizing patterns, and refining their strategies. A difficulty level that once felt appropriate may become too easy over time, resulting in boredom during key narrative or climactic moments.

A more flexible solution is needed: one that evolves alongside the player. DDA addresses this gap by responding in real-time to performance or even emotional state. Instead of locking players into a single challenge level, DDA systems personalize the experience moment to moment, helping maintain engagement across a broader spectrum of player ability.

4. Foundations of adaptive difficulty

While fixed difficulty settings offer a simple solution to balancing challenge, they often fall short in addressing the diverse and evolving needs of players. DDA has emerged as a promising alternative. However, its development is not just a technical shift. It is grounded in longstanding design practices and psychological theories. Understanding these foundations helps explain why adaptive systems are more than reactive mechanics; they are rooted in principles of engagement, learning, and motivation. This section examines the historical origins of adaptive design and the theoretical models, such as Flow Theory, that continue to influence the understanding and implementation of DDA today.

4.1 Flow theory in dynamic systems

One of the most influential frameworks behind DDA is psychologist Mihaly Csikszentmihalyi's Flow Theory [1]. Flow describes a mental state of deep immersion, marked by intense focus, a sense of control, and intrinsic enjoyment in the activity itself. In games, flow occurs when the challenge posed by the game is well-matched to the player's abilities, a balance that fosters engagement without tipping into boredom or anxiety.

According to Csikszentmihalyi, if a task is too easy for the player's current skill level, it results in apathy or boredom; if the task is too difficult, it induces worry, frustration, or even anxiety. Flow emerges in the narrow channel between these extremes, where players are stretched just enough to stay immersed and motivated. This balance is illustrated by the challenge-skill model in **Figure 1**, which visually represents how an optimal experience arises when task difficulty aligns with individual capability.

A well-tuned DDA system aims to preserve this equilibrium by continuously adapting gameplay conditions in real time, responding to fluctuations in player skill, focus, and emotional state. Rather than relying on static difficulty settings, such systems adjust challenge dynamically to keep the player within the optimal "flow zone."

This principle is foundational to both performance-based and emotion-based adaptive systems. The phenomenon of being "in the zone" reflects this state of flow, often characterized by heightened arousal, sustained concentration, and a diminished awareness of time and self. While flow is frequently associated with high arousal and positive valence, these are distinct emotional dimensions—arousal refers to the intensity of the emotional state, while valence indicates its pleasantness or unpleasantness. By supporting this optimal state through real-time adaptation, DDA can lead to more engaging, personalized gameplay experiences.

4.2 Beyond flow: Motivation, feedback, and personalization

Beyond Flow, adaptive gameplay connects to broader ideas in motivation, learning, and player psychology. Games are often described as "systems of feedback"—they reward progress, signal failure, and provide clear goals. When this feedback is timely, relevant, and matched to the player's actions and abilities, it promotes a satisfying loop of effort and reward.

Motivation theories such as Self-Determination Theory (SDT) highlight three psychological needs that drive engagement: autonomy, competence, and relatedness [3]. Adaptive difficulty directly supports the need for competence by adjusting challenges to align with player skill, helping players feel capable and successful. This reinforcement loop, where a player attempts a task, receives feedback, and either advances or tries again with better information, is essential for maintaining both engagement and learning.

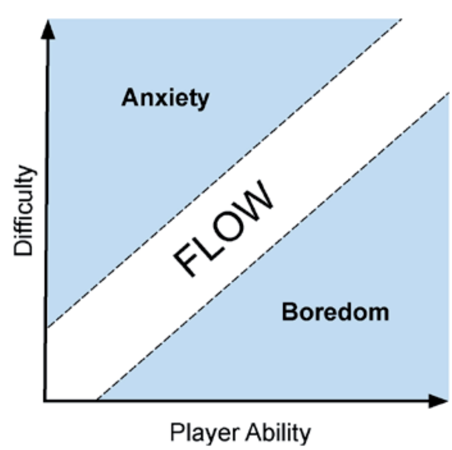

Figure 1.
Flow state is achieved when challenge matches player skill. Too much challenge causes anxiety; too little leads to boredom (Reproduced with permission from [2]).

This balance is not only motivational but also cognitive. According to Cognitive Load Theory [4], learners (and players) have a finite capacity for processing information. Adaptive difficulty can help regulate this load, reducing complexity when the player is overwhelmed and increasing it when the player is under-challenged, to ensure that the experience remains mentally manageable and engaging.

Feedback also plays a crucial role in how players perceive their own improvement. Systems that offer too little challenge risk making the player feel uninvested, while systems that are too punishing can lead to discouragement. Adaptive difficulty systems help maintain a motivational sweet spot, offering tasks that are just hard enough to promote effort, without overwhelming the player. This idea parallels scaffolding in educational design, where instruction adapts to a learner's needs to support growth.

Personalization amplifies these effects. In educational and entertainment contexts, research has shown that personalized systems—those that adapt to a user's preferences, pace, or performance—lead to better retention, deeper engagement, and longer play sessions. Adaptive systems that respond to emotional cues (e.g., frustration, boredom) or physiological signals (e.g., heart rate, gaze behavior) take this even further, potentially allowing for real-time difficulty tuning that meets the player exactly where they are.

By going beyond static levels and tailoring the experience to the individual, adaptive systems can meet the player's needs across multiple dimensions: cognitive, emotional, and motivational. This personalization helps create a gameplay experience that feels responsive, rewarding, and tailored to each player.

5. Dynamic difficulty adjustment types

DDA systems can be categorized by the types of data they use to assess the player's state and determine how to adjust gameplay. While the core goal remains the same—aligning challenge with player ability or experience—the methods used to achieve this vary widely. Some systems monitor performance metrics, while others assess emotional or physiological cues; many modern implementations combine multiple inputs to deliver more personalized and context-aware adjustments.

The following subsections outline three major types of DDA approaches: performance-based systems, emotion-based systems, and hybrid systems that combine both.

5.1 Performance-based systems

Performance-based DDA is the most common form of adaptive difficulty. These systems monitor the player's performance by tracking metrics such as accuracy, reaction time, win-loss ratio, and remaining health. Based on this data, the game dynamically adjusts the difficulty, for example, by increasing enemy speed, spawn rate, or AI behavior when the player is succeeding, or by reducing obstacles and offering assistance when they are struggling.

This approach is attractive because it directly responds to observable gameplay outcomes. It allows for real-time adjustment without requiring player input, and can be finely tuned to match specific design goals. Because performance is relatively easy to measure and interpret, these systems are among the most widely implemented in both commercial and experimental games.

Performance-based systems are especially well-suited to fast-paced or competitive games. A well-known commercial example is *Left 4 Dead*, where the game's AI Director adjusts enemy waves in real time based on team performance. Another frequently cited case is *Resident Evil 4*, which features a hidden difficulty adjustment system that modifies enemy behavior, damage, and item drops in response to player performance—creating a smoother experience without explicitly informing the player.

Academic research on performance-based DDA encompasses a range of genres and platforms. Early work explored dynamic challenge scaling in first-person shooters [5, 6], with subsequent studies focusing on balancing player experience through in-game performance metrics [7]. This approach has been extended to Massively Multiplayer Online environments through dynamic enemy tuning based on real-time data [8], and to the Multiplayer Online Battle Arena genre by adapting the challenge based on both team and individual performance [9].

More recent implementations have incorporated machine learning and novel interaction contexts. One study used deep learning to classify player skill and dynamically adjust difficulty in an arcade shooter, enhancing engagement and immersion [10]. Another application of DDA is to Virtual Reality (VR) exercise games, where procedurally generated environments are adapted to players' physical capabilities [11]. A separate model tailored difficulty in a 2D platformer using static danger zones and enemy behavior patterns [12].

Because performance metrics alone cannot capture the full range of player experience, researchers began exploring emotion-based systems—an approach intended to address not just external performance, but internal factors such as arousal, stress, and attentional engagement. While these systems are often described in affective terms, many rely on physiological indicators that primarily reflect emotional intensity (arousal), rather than emotional valence.

5.2 Emotion-based systems

While performance-based DDA adapts gameplay based on observable behavior, it does not always reflect how the player is feeling. A player may perform well but still experience stress, boredom, or cognitive overload. To address this limitation, emotion-based systems attempt to model the player's affective state and use that information to inform the adjustment of difficulty. These systems aim to support a more comprehensive understanding of the player's experience, capturing not only what the player is doing but also how they are emotionally responding to the game.

Emotion-based DDA systems rely on signals from the body to estimate how a player is feeling during gameplay. These systems use biosensors to detect emotional states such as stress, excitement, boredom, or fatigue. Common tools include heart rate monitors, skin conductance sensors, and brainwave measurement devices, such as electroencephalogram (EEG) headsets. However, most emotion-based systems focus on a specific signal to detect a single target emotion, such as using EEG to detect anxiety or heart rate variability (HRV) to infer stress. In practice, especially within affective computing and machine learning, these emotional states are often modeled as singular scalar values (e.g., "engagement" or "stress" scores from 0 to 100), rather than decomposed into independent dimensions like arousal and valence. These signals are then used to trigger adjustments in gameplay, such as modifying pacing, enemy density, or UI complexity. While these systems do not typically attempt to measure all emotional variables simultaneously, they can still offer targeted adaptation that enhances player comfort and engagement within specific gameplay contexts.

For example, researchers have used EEG to detect anxiety and mental focus in various types of games [13–15]. In contrast, others have explored excitement and arousal as key signals for adjusting difficulty in multiplayer settings [16]. Techniques like functional near-infrared spectroscopy have also been tested to measure mental workload [17], and HRV is often used to identify moments of stress or cognitive strain [18–21]. Skin conductance, measured through galvanic skin response, remains a popular method for tracking emotional arousal and has been widely applied in game adaptation research [22–24].

By gauging emotional cues such as stress, excitement, boredom, or fatigue, emotion-based systems can dynamically adjust the difficulty or gameplay elements in real-time. For instance, if the system detects signs of cognitive overload or frustration, it may lower the challenge or simplify interactions. If boredom or low arousal is detected, it may introduce new enemies, time pressure, or other stimulating events to counteract it. The goal is to create a more empathetic and responsive experience that adapts to both the player's internal state and their performance.

Although emotion-based DDA is still a growing area of research, a few commercial games have explored its potential. *Nevermind* uses heart rate sensors to detect player stress and dynamically adjusts the horror elements of its environment. *Deep*, a meditative VR experience, responds to diaphragmatic breathing to promote relaxation through gameplay. While not widespread in mainstream development, these games demonstrate how affective input can be used to shape both challenge and atmosphere, offering a glimpse into the future of emotionally adaptive design.

Despite its promise, emotion-based DDA presents challenges. Biosignals are often noisy, individual differences in emotional expression can be substantial, and interpreting affect in real-time remains a complex task. Moreover, while emotion-based systems offer a more human-centered lens on adaptation, they still represent only one side of what is required to sustain a flow state. These systems aim to support a more comprehensive understanding of the player's experience, capturing not only what the player is doing but also how they are emotionally responding to the game—typically through signals linked to emotional arousal rather than valence. Just as performance-based systems fall short by excluding emotion, emotion-based systems fall short by excluding ability. This limitation has led to a growing interest in hybrid approaches—systems that combine behavioral and physiological data to create a more comprehensive and adaptive gameplay experience.

5.3 Hybrid approaches

Hybrid DDA systems combine both behavioral and physiological data to inform adjustments to the difficulty level. By blending what the player is doing with how they might be feeling, these systems aim to form a more complete understanding of the player experience. Rather than relying on a single stream of input, such as performance metrics or emotional arousal alone, hybrid approaches integrate both to adapt gameplay in more nuanced and context-aware ways.

For example, a hybrid system might detect that a player is performing well, consistently landing shots, or completing objectives. Yet, physiological signals, such as elevated heart rate or skin conductance, indicate stress. Instead of raising the difficulty further, the system could maintain the current challenge level but ease the pressure by reducing time constraints or simplifying the interface. On the other hand, a player struggling with performance but showing signs of calmness or disengagement might trigger a gentle increase in challenge to re-engage their attention.

This dual-channel approach is particularly promising in the context of Flow Theory, which emphasizes the need for balance between challenge and skill, but also implicitly requires that players feel emotionally engaged, neither overwhelmed nor under-stimulated. By simultaneously accounting for both performance and affective cues, hybrid systems are well-positioned to maintain the conditions necessary for achieving a state of flow. In doing so, they can help sustain player engagement over longer sessions, support a wider range of player preferences, and create more meaningful, personalized experiences.

Hybrid systems are naturally more complex to develop. They require the fusion of multiple data streams, calibration across individuals, and often the use of machine learning models to classify affect and interpret gameplay context. Still, the payoff is a richer, more empathetic gameplay experience—one that responds to both what the player is capable of and how they are experiencing the game.

Recent studies have begun exploring these hybrid approaches in greater depth, combining performance data with physiological signals to refine adaptive difficulty techniques [2, 22, 25]. This integration aims to address the limitations of relying solely on one type of data, marking a promising direction for future research and application in the field. As the field continues to evolve, hybrid systems represent a compelling step toward more adaptive, emotionally aware, and skill-sensitive game experiences.

6. Case study: Exploring adaptive difficulty in a first-person shooter

Understanding the real-world challenges of adaptive difficulty systems requires examining how different adjustment strategies influence the player experience. This case study investigates a custom-built first-person shooter designed to compare five difficulty adjustment methods: Fixed Difficulty, Player-Selected Difficulty, Performance-Based Adaptation, Emotion-Based Adaptation, and a Hybrid approach. Rather than seeking to determine a superior method, the study explores how these strategies affect player engagement, enjoyment, perceived challenge, and emotional response. By analyzing gameplay metrics, physiological signals, and self-reported experience, the study offers insights into how adaptive systems can support stable and satisfying gameplay across diverse players (**Figure 2**).

6.1 Study overview

This study draws on previously published research [2] that examined the impact of dynamic difficulty adjustment on player experience in a custom-designed first-person shooter. Five difficulty adjustment strategies—Fixed, Player-Selected, Performance-Based, Emotion-Based, and Hybrid—were compared in terms of their effects on perceived difficulty, engagement, stress, excitement, and enjoyment.

Participants played under all five conditions, with gameplay order randomized to control for sequence effects. The study focused on whether different adaptive methods would meaningfully alter emotional responses linked to arousal, or maintain engagement across varying gameplay conditions.

6.2 Experimental setup

Thirty-one Participants completed preliminary questionnaires on gaming habits and immersive tendencies, then were fitted with electroencephalography and heart

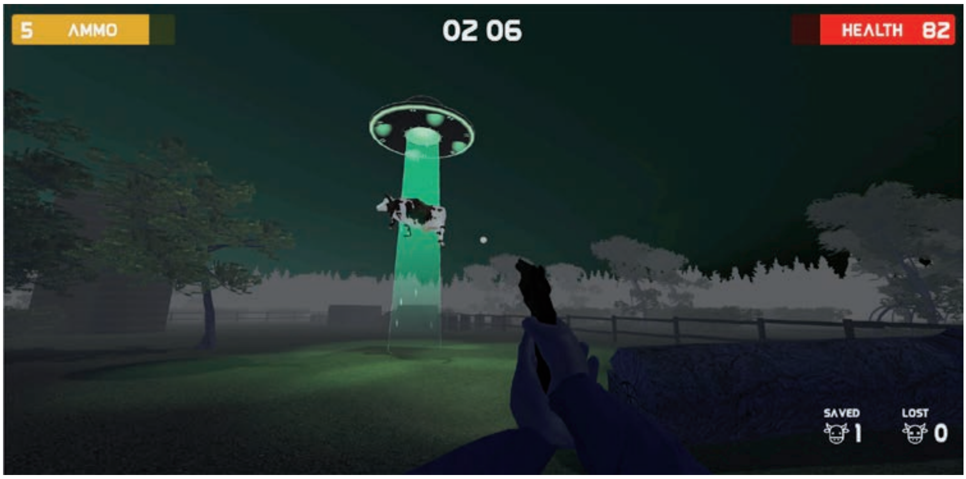

Figure 2.
Screenshot from the custom FPS game used in the study, designed to evaluate five different difficulty strategies. Reproduced with permission from [2].

rate sensors. After a brief training session to familiarize them with game controls, baseline physiological signals were recorded.

Each participant then played five short gameplay sessions, each using a different difficulty adjustment method. Sessions lasted up to 3 minutes or ended when the player's character was defeated. After each session, participants rated their perceived stress, focus, excitement, enjoyment, and difficulty. Throughout gameplay, performance metrics (e.g., accuracy, survival time) and physiological signals (e.g., engagement, stress) were continuously recorded.

For adaptive systems (Performance-Based, Emotion-Based, Hybrid), a 40-second moving average was used to smooth dynamic difficulty adjustments and avoid abrupt changes in challenge.

6.3 Difficulty adjustment methods

The five difficulty strategies varied in how they responded to player input, performance, or emotional state:

- *Fixed difficulty*: A constant, medium difficulty level served as a baseline comparison.

- *Player-selected*: Players chose their preferred difficulty (1–7) before starting. This allowed self-adjustment based on perceived skill or mood.

- *Performance-based*: Difficulty scaled using gameplay metrics such as accuracy, survival time, and objectives completed. High performance triggered increased challenge.

- *Emotion-based*: Difficulty adapted based on physiological signals. Affective data from EEG and heart rate sensors were used to maintain players in an optimal arousal range, aligned with Flow Theory, thereby avoiding boredom and overload.

- *Hybrid*: Combined both performance and emotional signals, with greater weight (0.6) given to emotion-based data, based on pilot testing that suggested emotional alignment had greater influence on player engagement.

6.4 Key findings

EEG-Derived Metrics. Physiological analysis focused on electroencephalography-derived metrics aligned with Flow Theory: Focus, Engagement, and Stress. Objective difficulty levels assigned by each system were also compared.

As shown in **Table 1**, mixed-effects modeling confirmed significant differences in assigned difficulty levels ($p < .001$), while experiential EEG metrics such as Focus, Engagement, and Stress remained largely stable across conditions. Although the model suggested subtle effects for some measures, none survived Bonferroni correction, reinforcing the interpretation that player engagement was broadly maintained across different adaptive strategies.

As shown in **Figure 3**, Emotion-Based adaptation consistently assigned higher difficulty levels (mean 5.93 on a 7-point scale), while Fixed and Performance-Based methods produced moderate difficulty. Player-selected conditions exhibited the greatest variability, reflecting individual differences in initial preference.

Self-Reported Experience. Participant self-reports mirrored objective difficulty trends. Stress and perceived difficulty ratings differed significantly across conditions, with higher difficulty corresponding to greater perceived stress. However, ratings of excitement, focus, and enjoyment remained statistically similar across methods (**Table 2**).

These patterns are further illustrated in **Figure 4**, which shows that the Emotion-Based adaptation was perceived as the most difficult on average, while Fixed and Performance-Based methods were rated as moderately challenging. Emotion-based adaptation was perceived as the most difficult (mean 5.34 out of 7), closely followed by the Hybrid approach. Fixed and Performance-Based strategies clustered around moderate difficulty ratings (approximately 4.0). Player-selected difficulty again showed the widest range, underscoring the variability in player self-assessment.

Overall, our results did not reveal one difficulty strategy as consistently superior in terms of player enjoyment, focus, or engagement. However, this lack of divergence may be meaningful in itself. Despite varying challenge levels—some systems adapting to stress, others to performance—players across all conditions reported similar

Metric	Mixed Model p -value
Difficulty	$< .001$
Focus	.034
Excitement	.144
Interest	.280
Engagement	.029
Stress	.040
Relaxation	.363

Table 1.
Summary of mixed-effects model results across difficulty adjustment methods. Difficulty reflects the actual in-game challenge level; all other metrics are derived from EEG signals. Only Difficulty remained significant after Bonferroni correction ($\alpha = .0071$).

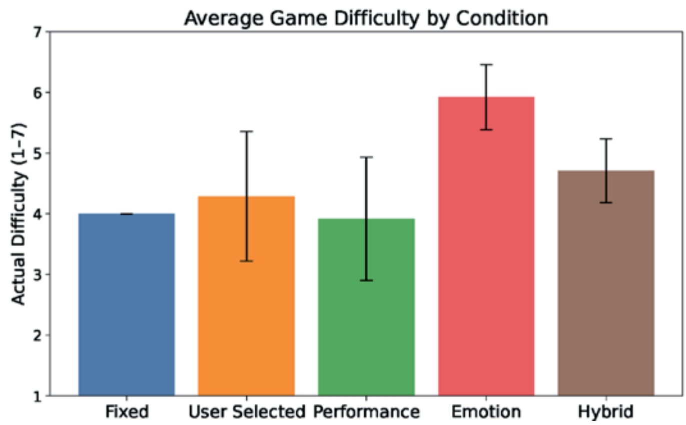

Figure 3.
Mean difficulty level assigned during gameplay, grouped by difficulty adjustment method. Bars represent the average setting applied by the system, with error bars showing standard deviation. The Emotion-Based system consistently assigned higher difficulty levels, while the Fixed and Performance-Based methods centered on moderate challenge. User-selected difficulty showed the widest variation across participants.

Question	Friedman Test
Please rate your average stress level while playing this iteration of the game	χ^2 (4) = 23.135, $p < .001$
Please rate your average excitement level while playing this iteration of the game	χ^2 (4) = 4.215, $p = .378$
Please rate your average focus level while playing this iteration of the game	χ^2 (4) = 2.675, $p = .614$
How much fun did you have playing the game this iteration?	χ^2 (4) = 4.098, $p = .393$
How difficult would you rate this iteration of the game?	χ^2 (4) = 21.766, $p < .001$

Table 2.
Friedman test results for selected post-questionnaire items comparing perceived experience across difficulty adjustment methods.

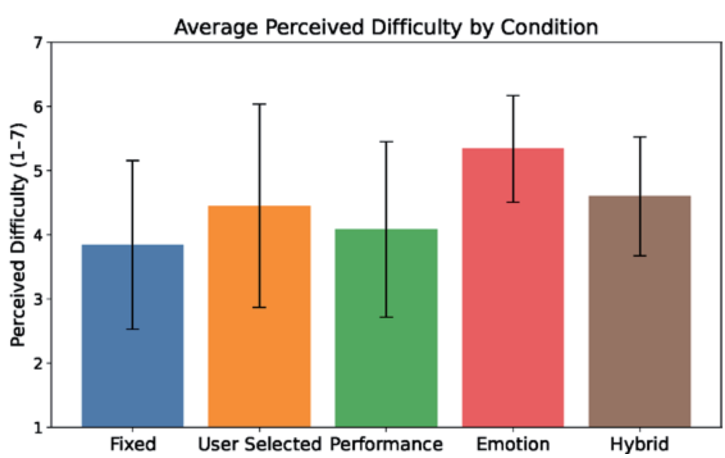

Figure 4.
Average perceived difficulty rating for each difficulty adjustment method, based on post-condition questionnaire responses. Bars represent the mean reported difficulty (1–7), with error bars indicating standard deviation. Participants rated the Emotion-Based condition as the most difficult on average, while Fixed and Performance conditions were perceived as more moderate. User-selected difficulty showed the greatest variability across participants.

emotional states based on Emotiv's predefined affective metrics. Both EEG signals and self-reports suggest that engagement was broadly sustained, regardless of how difficulty was tuned. This stability may reflect a core strength of adaptive systems: rather than amplifying emotional differences, they help ensure a consistent and playable experience across a diverse player base.

6.5 Design implications

Adaptive difficulty systems are often expected to produce highly personalized experiences; however, our findings suggest that their role may be more stabilizing than transformative. Across all adjustment strategies—whether based on performance, emotion, or a hybrid—players reported similar levels of enjoyment, focus, and excitement. This consistency highlights a key strength of adaptive design: its ability to maintain engagement across a diverse player base, even when gameplay conditions differ behind the scenes.

Rather than viewing this uniformity as a shortcoming, it may reflect successful alignment between challenge and ability. Flow Theory posits that engagement peaks when difficulty matches skill, and our results suggest that each system achieved this balance in its own way. Players remained emotionally invested, regardless of the method used to tune difficulty, suggesting that adaptation can support immersion without drawing attention to itself.

At the same time, our data challenges the assumption that all players want to be optimally challenged. The broad range of difficulty selections in the User-Selected condition points to varied motivations. Some participants may have been seeking mastery or competition, while others may have preferred a relaxed or exploratory experience. These findings reinforce the idea that adaptive systems should consider not just performance, but intent. Player-centered adaptation means supporting different play styles—not simply pushing everyone toward high difficulty, but tailoring the experience based on what the player wants out of the session.

Finally, the convergence across physiological signals, gameplay telemetry, and self-report lends confidence to these conclusions. While no single metric revealed dramatic differences, together they painted a coherent picture of stable engagement. This suggests that the value of adaptation may lie less in creating extreme highs and lows, and more in offering gameplay that remains steady, immersive, and responsive across a wide spectrum of player experiences.

7. Limitations of adaptive difficulty

Although adaptive systems can help stabilize player engagement across diverse gameplay conditions, they are not without tradeoffs. The same mechanisms that allow for smooth and responsive experiences can also create unintended consequences, particularly when players are unaware of, or uncomfortable with, how and when adaptation occurs.

One key concern is the impact on perceived competence. From the lens of Self-Determination Theory, players derive motivation from overcoming meaningful challenges. If a system quietly reduces difficulty to prevent failure, it may undermine the sense of achievement that comes from mastering a task. Even if players remain engaged, they may question whether their progress is truly earned, which can weaken the emotional payoff of success.

Similarly, adaptive systems that raise difficulty in lockstep with player improvement may obscure growth. Without visible markers of advancement, such as increased scores or tangible skill mastery, players may feel as if they are treading water, unable to track their development or feel a sense of momentum.

Transparency is especially critical in this context. While adaptive features are often framed as subtle or seamless, our findings suggest players are attuned to changes in difficulty, even when not explicitly informed. If those changes feel arbitrary or hidden, they risk undermining trust. This is particularly problematic in competitive or high-stakes settings, where fairness is of paramount importance. Perceptions of "invisible punishment," such as suspected skill-based matchmaking penalties, can damage player confidence even if the system is functioning as intended.

Lastly, many adaptive systems are guided by Flow Theory, which assumes players benefit most from maintaining a state of optimal challenge. But our findings and prior research suggest this model may not suit all playstyles. Some players engage for relaxation, while others do so for narrative, social connection, or curiosity. If systems rigidly aim to push players toward high challenge, they may misalign with those broader motivations.

These motivational mismatches, along with concerns about transparency and perceived competence, have behavioral consequences that extend beyond moment-to-moment experience. The Theory of Planned Behavior [26] suggests that players' intentions—and ultimately their behavior—are shaped by their attitudes toward the system, social context, and perceived control. If players feel that difficulty adjustments occur without their knowledge or input, or that they reflect negative evaluations of their skill, this can erode trust and reduce willingness to continue playing.

Beyond these experiential concerns, adaptive systems raise ethical questions. Because DDA can subtly influence player behavior, it could be exploited to serve design goals that benefit developers more than players—for instance, encouraging longer sessions or increased spending on microtransactions. When adaptation is used in this way without transparency or consent, it risks becoming manipulative rather than supportive.

Together, these limitations highlight the need for adaptive systems that go beyond performance metrics and skill matching. Truly player-centered adaptation requires not only flexibility and responsiveness, but also transparency, ethical boundaries, and an understanding of the diverse reasons players engage with games.

8. Conclusion

Dynamic difficulty adjustment offers significant promise for creating tailored gameplay experiences, but realizing that promise will require careful attention to both technical constraints and the evolving psychological expectations of modern players. However, despite its potential, several limitations still require attention for broader application. Technical constraints, such as the accuracy and reliability of physiological signals in emotion-based systems, present notable hurdles. Performance-based systems, while more straightforward, often neglect critical emotional and psychological dimensions of player experience, potentially limiting their ability to sustain engagement over time.

Hybrid systems, which integrate both physiological and performance metrics, align closely with principles from Flow Theory and offer a compelling direction for future research. Yet their practical application remains constrained by complexity, cost, and the challenge of achieving seamless, real-time integration of diverse data streams. In addition to technical hurdles, psychological factors must be considered carefully. Adaptive systems must preserve player autonomy, perceived fairness, and trust; transparent adaptation practices and user-centered design will be crucial for maintaining long-term satisfaction.

Looking forward, dynamic adaptation has the potential to evolve beyond simply adjusting difficulty along a traditional easy–medium–hard continuum. Rather than assuming that all players seek optimal challenge at all times, future systems could embrace adaptive profiles that align with the player's intent for the gaming session. Options such as "challenge-focused," "relaxation-focused," or "story-driven" experiences would allow players to express their goals more directly, with adaptation tuning itself accordingly.

To truly align with players' goals and expectations, future adaptive systems must account not only for real-time data but also for the psychological factors that drive continued engagement. As discussed earlier, behavioral models such as the Theory of Planned Behavior highlight the importance of perceived control and player attitudes—factors that are essential to fostering trust and long-term satisfaction. By considering not just performance metrics but also behavioral intentions and player experience, dynamic systems can overcome the limitations not only of traditional difficulty models but even of Flow Theory itself, recognizing that sustaining engagement sometimes means challenging the player, and other times means supporting exploration, immersion, or rest.

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Conflict of interest

The authors declare no conflict of interest.

Abbreviations

2D	two-dimensional
3D	three-dimensional
DDA	dynamic difficulty adjustment
EEG	electroencephalography
FPS	first-person shooter
HRV	heart rate variability
SDT	self-determination theory
UI	user interface
VR	virtual reality

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